

Light as the boundary dynamics of the field and origin of matter, inertia and gravity (UQSH)

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Abstract

Within the framework of the Universal Quantum Foam Hypothesis, this paper interprets light not as an object in space, but as the immediate limiting dynamics of an active field. It draws on ideas developed in the author's earlier publications and synthesizes them in a condensed form. The speed of light appears as the ontological limiting rate of all change, simultaneously the minimum and maximum absolute velocity. Matter is understood as the closed form of this limiting dynamics,

and inertia as the resistance of self-sustaining orbital structures to external phase shifts. Gravity arises as a field response to such self-bindings. In extreme field states, this dynamic leads to phase transitions of the medium without breaking the invariance of c . The speed of light remains constant across scales, while only the time rate of the respective regimes differs. Scale relativity is thus understood as the relativity of the form of time while the ontological dynamics remain constant. Light appears not as one phenomenon among others, but as the velocity of the universe itself.

1 Speed of light as field motion

The UQSH understands physical space not as a passively given background, but as an active, continuous field whose internal dynamics form the basis of all phenomena. In this sense, space is not the vessel of physics, but its carrier. Within this framework, light is not an object moving through a predetermined space, but rather the directly realized boundary dynamics of this field itself. It is not an entity within the field, but the elementary form of its movement.

The speed of light, c , is therefore neither merely an empirically determined constant nor simply an upper limit for motion, but rather the ontological limiting rate of all change. It denotes the maximum reorganization capacity of the medium itself. Simultaneously, it is both the minimum and maximum absolute speed. Minimal insofar as free, unbounded existence is only possible within this limiting dynamic, and maximum insofar as no field rearrangement, no information, and no interaction can occur faster than the field restructures itself. Below this limiting rate, bound forms arise; above it, no physically realizable dynamic exists.

Light is therefore not movement within the field, but rather movement of the field in its most open form. Along light's own worldline, neither time nor space exists, since its proper time is zero. For light itself, there is no duration, no path, and no sequence. All observed delays, stretches, and dilations arise solely from the perspective of bound structures within the same field regime. They are not properties of light, but properties of the medium and the observers organized within it.

It is not the light that adapts to the field, but rather the field that defines, in every state, what is possible as movement. Light always realizes precisely this maximum possibility. Its invariance is therefore not a special property of a particle, but an expression of the constancy of the field's internal boundary dynamics itself. In this sense, light is not one phenomenon among others, but the direct expression of the universe's capacity for movement.

2 Matter, inertia, and gravity as bound light

In this picture, matter does not appear as subcritically slow motion, but rather as closed form of boundary dynamics. An electron can be understood as a self-sustaining vortex of light whose internal circulation continues at the limit rate c , but in a cyclic and topologically bound form. Inertia does not arise from a lack of motion, but from the complete utilization of the internal dynamics of such structures. Any external acceleration interferes with an already maximally running cycle and necessitates a phase shift of the entire self-binding. Mass is thus the manifestation of the resistance of closed form of boundary dynamics to external reorganization.

Gravity and inertia are not separate phenomena in this context. Every self-sustaining motion constantly reorganizes the surrounding field, thereby creating a local density of the medium. This field response is gravity.

In bound patterns, it arises from the intrinsic rotation itself. The orbital motion that maintains a pattern generates an internal gravity because each phase can only exist through the continuation of the orbit. Inertia is thus the local manifestation of this internal gravity. It is not passive, but rather the dynamic consequence of motion sustaining itself.

This internal gravitational, however, does not exist in isolation. Bound patterns are simultaneously embedded in a larger field structure, whose condensations and stress gradients act upon them. In this sense, external gravitational force is the field-wide projection of the same dynamic that appears as inertia within a pattern. When a bound pattern encounters an externally curved field, its own self-binding couples to the large-scale field structure. The pattern does not follow a "force" but moves along the field organization that least reorganizes its internal dynamic. Free fall is therefore not a state of being driven, but rather the correspondence of its own orbital structure with the local field geometry.

Internal inertia and external gravity are thus two perspectives of the same principle: Inertia describes the resistance of a closed boundary dynamic to internal phase shifts, Gravity its embedding in the field-wide organization of the same dynamic.

Both stem from the same cause: motion sustains itself and, in doing so, shapes the field in which it exists. This self-sustaining mechanism is recursively nested. Each revolution, in turn, consists of finer revolutions, which themselves generate inertia and field density. Every stable structure is therefore a vortex of vortices, a hierarchically layered gravitational core within itself and simultaneously part of a larger field curvature.

3 The event horizon as a field-phase transition

In regions of extreme field saturation, such as near an event horizon, it is not the light itself, but the field that transitions into a different state of organization. This field-phase transition does not mark the exceeding or falling below the speed of light, but rather the transition from propagatable to completely internally bound dynamics. The boundary motion of light remains unchanged, but the medium increasingly loses the ability to directly propagate this dynamic outward. What appears from the outside as a slowing, stretching, or freezing of the motion is not a change in the light, but rather the growing inertia of the field in transmitting its own boundary dynamics.

Light is not movement through a medium, but rather a form of field movement itself. In an elastic field, a local excitation can immediately propagate to neighboring areas. This immediate transmission is what is experienced as light propagation. In a highly saturated field, however, this reorganization is increasingly inhibited. Each new excitation increases the internal tension, and the same boundary dynamic requires ever more field work to be transformed back into a propagating state. For light itself, there is no waiting, no duration, no delay, since it has no proper time. The entire experience of time dilation lies within the field and in the perspective of bound structures.

The boundary motion is not halted in such regions, but rather forced into internal cycles. It remains fully intact, but is no longer open, instead becoming topologically bound. Only when field states are reached that again possess sufficient elasticity can this bound dynamic be released. For external observers, this process appears as an extreme temporal stretching, while the released radiation manifests as a high-frequency form, since the

previously internally contained boundary dynamic is transformed into a highly condensed reorganization. The event horizon is thus not a point where motion ends, but a threshold at which boundary dynamics fully transition into internal field processes. The invariance of c is preserved because it is not a property of light, but a property of the field itself.

4 C as the universal reference point of all scales

This invariance is effective across scales. The speed of light cannot change, neither in our experiential world nor in subsequent or superordinate regimes, because it does not belong to a specific level but represents the limiting rate of the field itself. It is not a property of a scale but the common dynamic basis of all scales. What differs between the levels is therefore not c , but the respective time rate of the regime, that is, the internal rhythm at which processes unfold and causality is experienced. Each scale possesses its own rhythmic organization of its processes and thus its own form of experiencing duration, sequence, and simultaneity.

A process that appears instantaneous on one scale can represent an extended dynamic on a finer level, while conversely, processes on a finer scale can appear practically timeless on a coarser one. Causality itself remains unbroken, but its temporal form is relative to the respective level of organization. Scale relativity therefore does not mean the relativization of the laws of nature, but rather the relativization of the form of time while the ontological dynamic remains constant.

The scales continue indefinitely, in principle, because the field knows no privileged resolution. There is neither a final subtlety nor a definitive overarching order, but rather an open hierarchy of regimes, all connected by the same boundary dynamic. The speed of light functions as a uni-

versal reference point between all levels of being. It couples the scales without adapting to them, guaranteeing a common physical order despite differing temporal experiences. In this sense, light is not a phenomenon among others, but the speed of motion of the universe itself, the invariant beat to which every form of reality is aligned.

5 Final sentence for this scale

In conclusion, this ontological framework does not represent a closed worldview, but an open space for interpretation. The Universal Quantum Foam Hypothesis is not intended as a replacement for existing theories, but as an overarching perspective in which diverse physical descriptions can find common ground. It does not prescribe definitive forms, but describes principles of organization from which a wide variety of structures can be derived.

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